

Vector Similarity Handbook

Structural and semantic similarity of hidden representations: every term, every diagnostic, worked numbers from this project, and a reusable test battery (notebook V1)

1. Two kinds of similarity, one page

Structural similarity asks about the *shape* of a representation space, or the agreement of two shapes: how many directions carry variance, whether two models organize the same inputs the same way, what transformation separates their coordinate systems. Its tools are spectra, CKA, Procrustes, fitted linear maps. **Semantic similarity** asks about *individual vectors*: does this query point the same way as that item, is this photo already indexed, which caption matches. Its tools are cosine, R@K, membership gaps. The two meet in one fact this project measured end to end: when structural similarity holds between two encoders (same geography, different coordinates), one matrix makes their semantic similarities interchangeable - a text query scores a phone-encoded photo and its server twin at 0.100 vs 0.122, near-identically, through a 0.79 MB adapter.

2. The primitive measures

Term	Question it answers	Definition / example
mean	how large on average?	sum / count. Row means of a similarity matrix reveal hubness: a layer generically similar to everything.
max	what is the largest value?	-
argmax	WHERE is the largest value?	<code>np.argmax(row)</code> = index of the best match. Per-row argmax of a CKA matrix is the measured layer correspondence; <code>[0,1,2,...]</code> = diagonal, repeated values = hub columns.
magnitude $ v $	how long is the vector?	sqrt of sum of squares. (3,4) has $ v = 5$; (6,8) has 10 - same direction, twice the length.
L2 normalization	remove length, keep direction	$v / v $. (3,4)/5 = (6,8)/10 = (0.6, 0.8) - after it, only angle can differ, and dot product = cosine.
cosine	do two vectors point the same way?	$u \cdot y / (u y)$, in $[-1, 1]$. Blind to length. The metric every vector index computes.
R@K	is the right item in the top K?	fraction of queries whose true target ranks in the first K. Gallery-size dependent: $R@1 = 0.557$ on 1,000 candidates. Chance = $1/\text{gallery}$.
singular values	the volume knobs of a matrix	$W = U \cdot S \cdot V^T$; S holds one scale factor per direction. Rotation = all knobs at 1. The shipped adapter's knobs span 0.052-3.592 (69x).

3. Comparing whole spaces

3.1 CKA - relational structure, no coordinates

Build each model's $n \times n$ similarity table over the same inputs; CKA is the cosine between the two centered tables:

$$\text{CKA}(X, Y) = |Y^T X|_F^2 / (|X^T X|_F - |Y^T Y|_F)$$

Invariant to rotation, permutation and uniform scaling - exactly the symmetries that leave a network's function unchanged - but NOT to anisotropy differences, which is why a modest CKA can sit beside a strong ridge fit without contradiction. Measured here: trained pair 0.359 mean vs random control 0.068 (5.3x); diagonal strictly monotone L3-L11 (Spearman rho = 1.000).

3.2 Procrustes - the rigid hypothesis

$$R^* = \text{argmin over orthogonal } R \text{ of } |X - RY|_F^2; \text{ solution: } X^T Y = U - S - V^T, R = U - V^T$$

Orthogonality preserves all distances and angles: rotation and reflection only, every volume knob welded to 1. A good Procrustes fit is therefore strong evidence of rigidly congruent geometry. Its failure is equally informative: here R-squared 0.038 while its output's *cosine* to target is 0.775 - directions substantially right, volumes badly wrong. That pair of numbers is the anisotropy diagnosis in miniature.

3.3 Ridge / stitching - the working hypothesis

$$W = (X^T X + aI)^{-1} X^T Y$$

Full linear freedom: rotation + per-direction scaling + shear. Closed form, deterministic, decomposes into one regression per output dimension sharing one inverse - so the sample requirement is rows per INPUT dimension (≥ 5). Held-out R-squared is the score; a shuffle control (deliberately misaligned rows) is the power check. Measured: 0.592 (vision pair), 0.737-0.741 mean (LLM pair), shuffle controls at ~ 0 or negative.

3.4 The ladder of strictness

Method	Allowed transformation	A pass means
cosine on raw vectors	none	same coordinates - almost never true across models (measured -0.003)
Procrustes	rotation / reflection	rigidly congruent geometry (failed here: anisotropy)
ridge / linear map	rotate + scale + shear	same geography, different coordinates AND volumes (passed: 0.592 / 0.737)
MLP / nonlinear	almost anything	if it beats linear, the relationship is genuinely nonlinear (here +0.003: it is not)

4. The anisotropy battery - six diagnostics for one space

Anisotropy: variance concentrated in few directions. It inflates similarity scores (unrelated vectors share the dominant component), promotes hubness, defeats rotation-only alignment, and biases variance-weighted metrics. Six tests, each cheap:

#	Diagnostic	Formula / recipe	Reference points
1	mean random-pair cosine	average cos over random pairs of L2-normalized rows	isotropic ~ 0.00 ; MobileCLIP here $+0.18$; strongly anisotropic >0.5
2	mean-direction ratio	$ \text{mean vector} / \text{mean } \text{row} $	~ 0 isotropic; large = shared dominant direction
3	PC1 / top-k variance share	PCA on centered rows	PC1 $\sim 1/d$ isotropic; MobileCLIP 0.054; >0.3 concentrated
4	rank-1 energy	$s_1^2 / \sum s_j^2$ of the centered SVD	the CKA matrix here: 97% - hubness-dominated
5	effective rank	$\exp(\text{entropy of the normalized spectrum})$; or participation ratio $(\sum e)^2 / \sum e^2$	MobileCLIP 141 of 512; GPT-2 layers 2.2-10.9 of 768 (rogue dimensions); Pythia 9.3-92.7
6	survival under correction	center, then remove top-k PCs, recompute the finding	survives = real; vanishes = it was the shared component

Anisotropy vs hubness. Anisotropy is a property of the space (directional concentration); hubness is its symptom in similarity tables (some items become everyone's neighbour). The chain: shared dominant directions make many vectors look broadly similar, so a few rows/columns light up everywhere. Diagnose hubness by row/column means and the rank-1 share; correct it by dividing out marginals (lift) - which here raised within-two-layers correspondence from 62% to 77%.

5. Semantic similarity in production

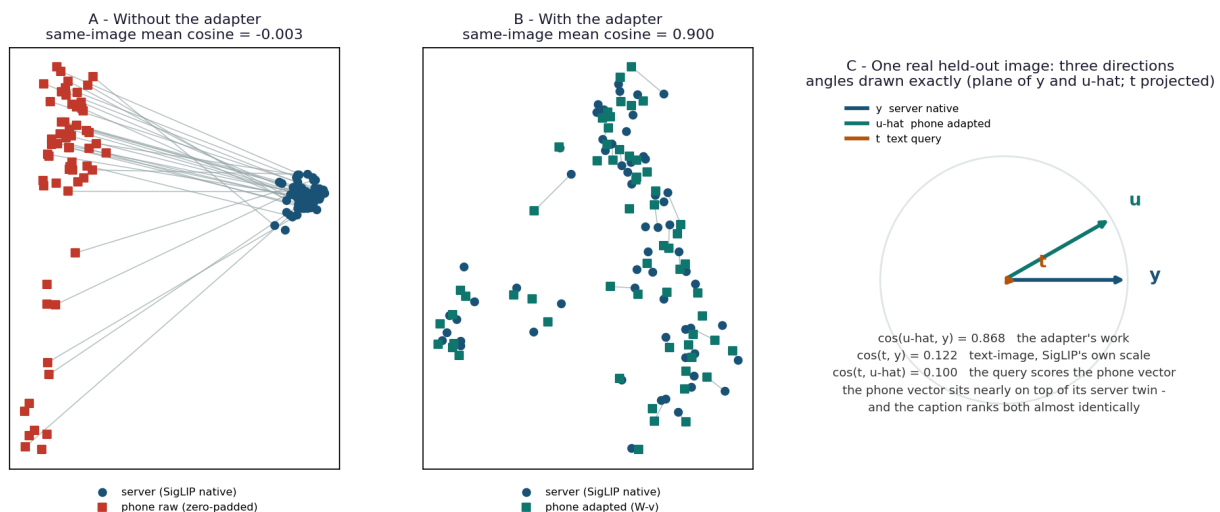


Figure 1: the project's own evidence, real held-out embeddings. Structure (panels A-B): one linear map moves cross-encoder same-image cosine from -0.003 to 0.900. Semantics (panel C): a text query then scores the phone vector and the server vector near-identically - retrieval, dedup and membership follow.

Semantic operation	Mechanism	Measured here
text -> image search	$\cos(t, u\text{-hat})$ ranking	$R@5 = 95.1\%$ of ceiling across tiers
image -> image similarity	cos between stored vectors	image-as-query $R@1 = 0.987$
membership / dedup	true-twin cosine minus best impostor	gap $+0.075$ mean, positive for 98.7%
negative control	raw vectors, no adapter	0.001 = exact chance - the capability lives in W

6. Benefits, where this is found, implications

6.1 What structural similarity buys

Interoperability without retraining - two encoders, one index, one seconds-cheap matrix; refit on upgrade instead of re-indexing (this project's deliverable). **Model stitching and reuse** - layers or components of independently trained models connected through small maps. **Compression insight** - effective rank tells you how many dimensions are truly used; quantization and truncation decisions follow. **Measurement hygiene** - anisotropy diagnostics separate real findings from shared-component artifacts before you publish them.

6.2 What semantic similarity buys

Retrieval (search, RAG grounding), **deduplication and membership** (the gap statistic), **clustering and recommendation**, **drift monitoring** - the anisotropy battery run periodically on production embeddings detects collapse or distribution shift cheaply, before quality metrics notice.

6.3 Where the phenomena appear

Anisotropy and rogue dimensions: documented across LLM hidden states (measured here at effective dimensionality 2.2-10.9 for GPT-2), sentence embeddings, and contrastive vision spaces (MobileCLIP 141/512). Hubness: any high-dimensional k-NN system. Convergent structure: between independently trained models of the same modality (this project), and - per the Platonic Representation Hypothesis - increasingly across modalities as models strengthen; this project is two consistent points on that curve, not its proof.

6.4 Implications and cautions

Embeddings are content-bearing. If one matrix translates between spaces, an address in any convergent space is close to an address in all of them; treat stored vectors with the same care as the data they encode (inversion models recover approximate content). **Similarity scores are not absolute.** SigLIP's sigmoid training keeps true-match text-image cosines near 0.15; thresholds must be calibrated per space, and cross-space numbers compared only after alignment. **Metrics encode assumptions.** Variance weighting imported a 0.097 phantom asymmetry here; R-squared punished magnitude the index discards. Choose the metric the system actually uses. **Negative results need power checks first** - rows per input dimension, the shuffle test, artifact sanity - before 'no similarity' is believed.

6.5 The reusable battery

Notebook V1_vector_diagnostics runs all of the above on any $[n, d]$ array (and the cross-space + semantic blocks on any paired second array): sanity, the six anisotropy diagnostics with reference points, correction-survival, ridge-vs-Procrustes with the shuffle control, volume knobs, R@K and the membership gap. Verified against this project's artifacts: it reproduces R-squared 0.598 / cosine 0.902 on a fresh random split of the vision pairs.